MILITARY SPECIFICATION

ALARM UNIT, CHEMICAL AGENT, AUTOMATIC ALARM, ABCA-M42

This amendment forms a part of Military Specification MIL-A-51384A(EA), dated 5 November 1979, and is approved for use by all Departments of the Department of Defense.

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2.1: Under "SPECIFICATIONS" "FEDERAL" add "PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes".

2.1: After "SPECIFICATIONS" "FEDERAL" add:

"MILITARY"

"MIL-P-60312 - Parts, Molded, Plastic Foam, Polystyrene (For Use With Ammunition)."

2.1: Under "DRAWINGS" add:

"D5-15-8754 - Overpack Alarm Unit, Chemical Agent Automatic Alarm, M42."

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3.7: Delete in its entirety and substitute the following:

"3.7 Top panel and gasket leakage. There shall be no evidence of air leakage through the top panel or gasket of the alarm unit or through or around any of the components mounted thereon when the internal air pressure of the alarm unit is increased to 8.0 ± 0.5 inches (20.32 ± 1.27 cm) of water greater than the ambient pressure and the alarm unit is immersed in a tank of water to a maximum depth of 2.0 inches (5.08 cm) measured from the top surface of the top panel to the surface of the water for at least 45 seconds when tested as specified in 4.4.4.5."

3.8 Delete in its entirety.

3.11: Delete "The accumulation of water within the case of the alarm unit shall not be a cause for rejection."

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4.4.3.3 Delete:

"103 Top panel leakage 4.4.4.5,
104 Gasket leakage 4.4.4.6", and substitute
"103 Top panel and gasket leakage 4.4.4.5."

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4.4.4.5: Delete in its entirety and substitute the following:

"4.4.4.5 Top panel and gasket leakage. Install the alarm unit top panel and gasket onto a test housing conforming to Drawing D5-15-4824. The test housing shall be fitted with a suitable air inlet connection attached to a magnehelic gage, having an accuracy of not less than 2.0 percent full scale deflection in 0.5 inches of water increments, which is then connected to a regulated pneumatic source. Increase the air pressure within the alarm unit until 8.0 ± 0.5 inches (20.32 ± 1.27 cm) of water is indicated on the magnehelic gage. Immerse the alarm unit in a tank of water to the specified depth for the required period of time. The water should contain a suitable wetting agent (e.g., 1 gram of Aerosol OT per 500 milliliters of water). Examine the top panel and gasket for evidence of leakage as required by 3.7. A leak is defined as a bubble making or breaking from the surface of the alarm unit top panel or gasket.

4.4.4.6: Delete in its entirety.

Under "5. PACKAGING" add:

"5.1.1 Moisture content. The moisture content of the molded polystyrene overpack (D5-15-8754) at the time of unit packing (5.1) shall not exceed 0.4 percent when tested in accordance with MIL-P-60312. The rejected overpacks shall be redried and retested to these requirements."

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5.2.1(2) line 3: Delete "11-1/8 by 8-3/8 by 6-3/4" and substitute "10-15/16 by 7-5/8 by 5-3/8."

5.2.1(2) line 5: After "respectively." insert "Fiberboard filler pads, conforming to class domestic, variety SW, grade optional of PPP-F-320, shall be included as necessary to ensure a tight pack."

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